

Exp 3	Performance Analysis of IIR Filters
Date	3.1 Design and Comparative Analysis of Butterworth and Chebyshev Type-I IIR Band-Pass Filters Using MATLAB

Code:

```
clc; clear; close all;
```

```
% Specifications
```

```
Fs = 2000;
```

```
Rp = 1;
```

```
Rs = 40;
```

```
Wp = [300 600]/(Fs/2);
```

```
Ws = [200 700]/(Fs/2);
```

```
% Butterworth Filter
```

```
[n1,W1] = buttord(Wp, Ws, Rp, Rs);
```

```
[b1,a1] = butter(n1, W1, 'bandpass');
```

```
% Chebyshev Type-I Filter
```

```
[n2,W2] = cheblord(Wp, Ws, Rp, Rs);
```

```
[b2,a2] = cheby1(n2, Rp, W2, 'bandpass');
```

```
% Magnitude Response
```

```
figure;
```

```
freqz(b1,a1);
```

```
title('Butterworth Filter');
```

```
figure;
```

```
freqz(b2,a2);
```

```
title('Chebyshev Type-I Filter');
```

```

% Group Delay

figure;

grpdelay(b1,a1); hold on;

grpdelay(b2,a2);

legend('Butterworth','Chebyshev-I');

title('Group Delay Comparison');


% Pole-Zero Plot

figure;

subplot(1,2,1);

zplane(b1,a1);

title('Butterworth');

subplot(1,2,2);

zplane(b2,a2);

title('Chebyshev-I');

% Impulse Response

figure;

subplot(2,1,1);

impz(b1,a1);

title('Butterworth Impulse Response');

subplot(2,1,2);

impz(b2,a2);

title('Chebyshev-I Impulse Response');

% Pole Magnitude

figure;

subplot(1,2,1);

stem(abs(roots(a1)),'filled');

title('Butterworth Pole Magnitude');

subplot(1,2,2);

stem(abs(roots(a2)),'filled');

title('Chebyshev Pole Magnitude');

```

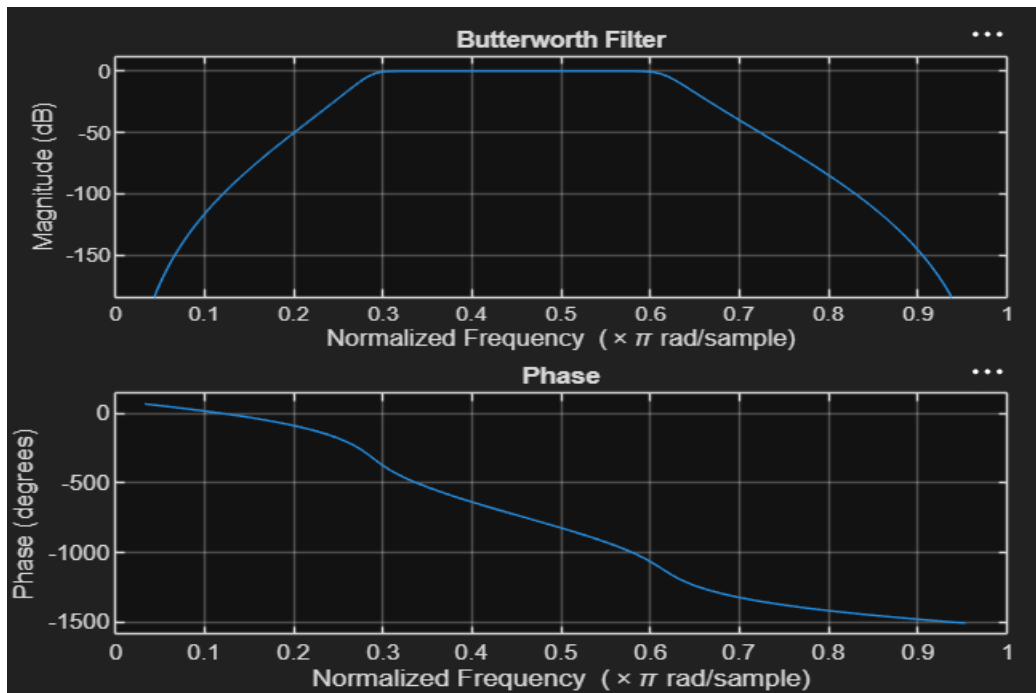
% Display Filter Orders

```
disp(['Butterworth Order = ', num2str(n1)]);
```

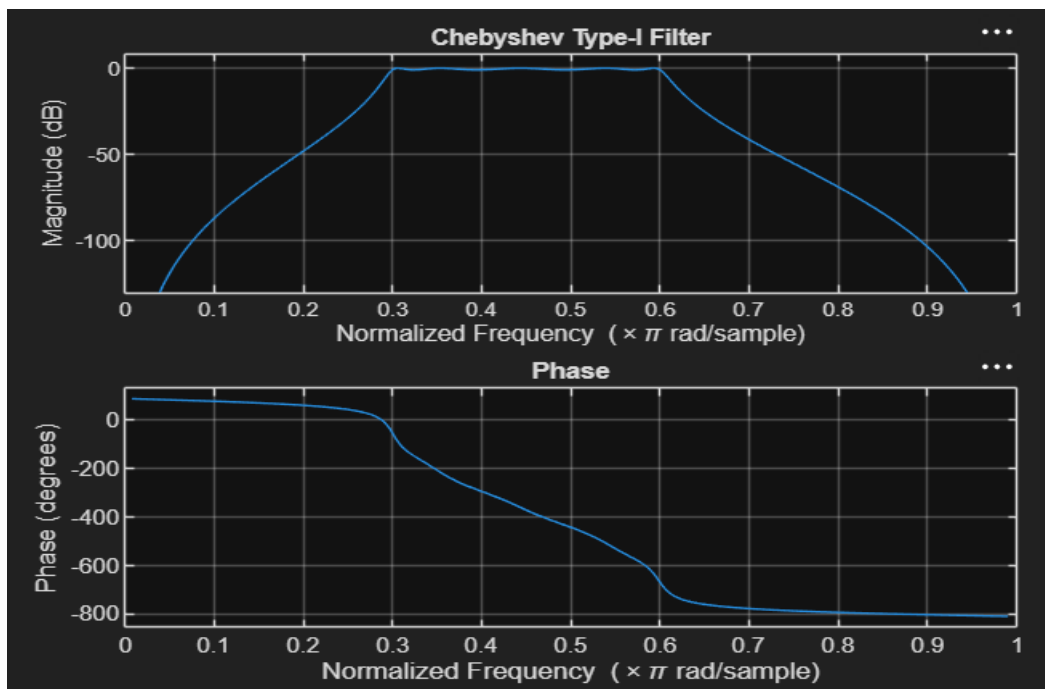
```
disp(['Chebyshev Order = ', num2str(n2)]);
```

Output:

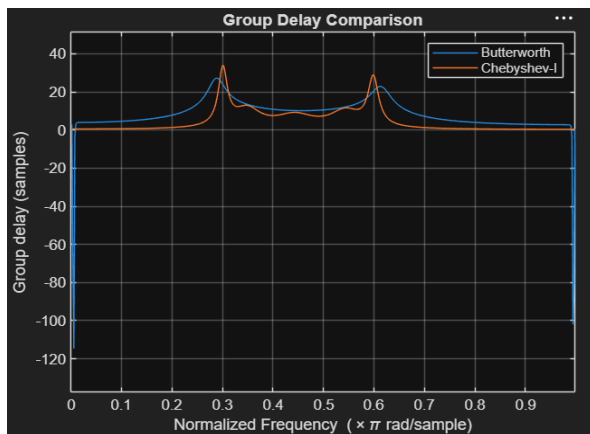
Objective 1:



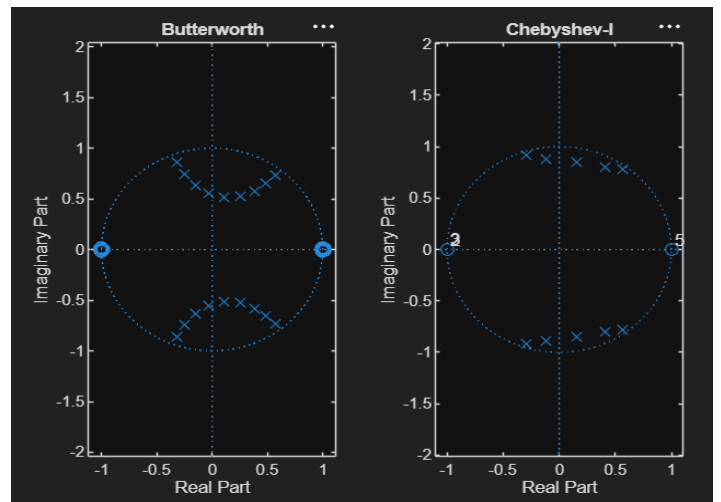
Objective 2:



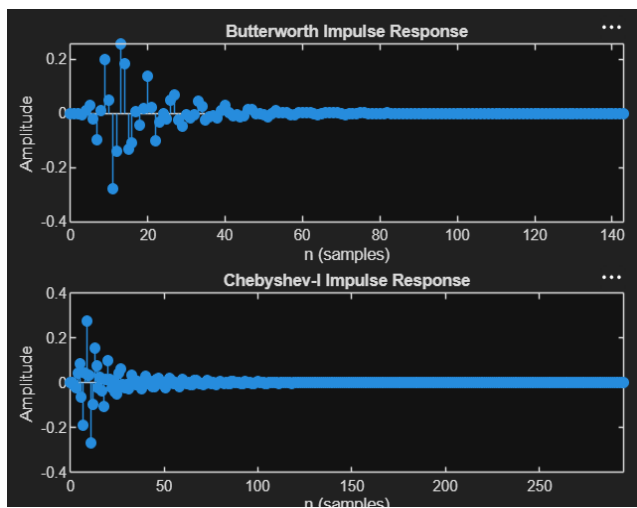
Objective 3:



Objective 4:



Objective 5:



Objective 6:

